



Adiel González

Introduction

Contents

Introduction	i
1 Logic	1
1.1 Formal languages	2
1.2 Propositional logic	3
1.3 First order logic	4
1.4 Non-classical logic	5
2 Introduction to set theory	7
2.1 Elements of set theory	8
2.2 Relations and functions	9
2.3 Natural numbers	10
2.4 Integers and rationals	11
2.5 Real numbers	12
2.6 Ordinals	13
2.7 Cardinality	14
2.8 Choice axiom	15
2.9 Well foundedness	16
3 More set theory	17
3.1 Classes and sets	18
3.2 Limits of infinity	19
3.3 Universes	20
3.4 Constructibility	21
3.5 Further topics on set theory	22
4 Structures	23
4.1 Historical note	24
4.2 Sets as the foundation	25
4.3 Universal algebra	26
4.4 Introduction to categories	27
5 Foundations of mathematics	29
5.1 Category theory	30
5.2 Metamathematics	31
5.3 Mathematical logic	32
5.4 Type theory	33
5.5 Philosophical concerns	34

1

Logic

1.1 Formal languages

Every definition given here might be understood as not strictly mathematical. That meaning, we won't dive into details nor ontological remarks about the things that we define. Collections are to be understood in the intuitive sense, as well as known collections like the natural numbers. The natural numbers are given with their operations.

Definition. A *first order formal language* $\mathcal{L}$ is a collection of symbols, each identified with a type¹. It can be interpreted as a collection of collections, each of them being numerable².

- Constant symbols: $c_1, c_2, c_3, \dots$
- Variables: $x_1, x_2, x_3, \dots$
- Predicate symbols: $P_1, P_2, P_3, \dots$
- Relation symbols: $R_1, R_2, R_3, \dots$
- Function symbols: $f_1, f_2, f_3, \dots$
- Logical connectives: $\rightarrow, \neg, \vee, \wedge, \leftrightarrow$.
- Quantifiers: $\forall, \exists$.
- Parenthesis: $(,)$.

More generally, each relation symbol should have associated to it an “ n -arity”, that is, they relate n things, in which case each relation symbol that is m -ary should be denoted as R_n^m for some naturals m and n . We work under the assumption that these relation symbols are binary.

Definition. For a formal language $\mathcal{L}$, a *chain* is the juxtaposition of symbols of $\mathcal{L}$. More mathematically, a chain is a sequence (finite or infinite) of elements of $\mathcal{L}$.

Examples of chains are $x_1x_2x_3\neg, \neg\wedge x_1((((\text{and } (x_{43}P_{111}\wedge(f_{990}))\rightarrow)P_1R_1R_1f_2\wedge$. As you may see, these chains seem like nonsense. And they are. We shall give formations rules for what we want to be well formed chains, that is terms and formulas.

¹As in, this object is of type Natural number, or of type Logical connective.

²Countable. That is, to be able to make a one-to-one correspondence with, at most, all natural numbers.

1.2 Propositional logic

1.3 First order logic

1.4 Non-classical logic

First order logic as just presented, is also called classical logic. More specifically, it's first order monadic predicate logic. Now, this section treats some non-classical logics, mainly intuitionistic and modal logic. The use of these logics has interesting consequences for mathematics and its foundations.

2

Introduction to set theory

2.1 Elements of set theory

2.2 Relations and functions

2.3 Natural numbers

2.4 Integers and rationals

2.5 Real numbers

2.6 Ordinals

2.7 Cardinality

2.8 Choice axiom

2.9 Well foundedness

3

More set theory

3.1 Classes and sets

3.2 Limits of infinity

3.3 Universes

3.4 Constructibility

3.5 Further topics on set theory

4

Structures

4.1 Historical note

4.2 Sets as the foundation

4.3 Universal algebra

4.4 Introduction to categories

5

Foundations of mathematics

5.1 Category theory

5.2 Metamathematics

5.3 Mathematical logic

5.4 Type theory

5.5 Philosophical concerns